

Prompt Engineering

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Week 11: Domain-Specific Applications

Why Domain Context Matters

The "Last Mile" Problem

Generic LLMs (GPT-4, Claude) are trained on the "average" internet. They get you 80% of the way there. The last 20%—the domain specificity—is where value and risk live.

- **Vocabulary:** "Depression" means something different in Geology vs. Psychology vs. Economics.
- **Risk Tolerance:** A hallucination in a creative writing prompt is a feature. In a medical diagnosis, it's a lawsuit.
- **Workflow:** Professionals don't just want "answers"; they want outputs that fit their existing processes (e.g., SOAP notes, legal briefs).

Weekly Learning Objectives

- **Translate** domain expertise into prompt instructions and guardrails.
- **Build** knowledge packs, glossaries, and canonical examples (Few-Shot).
- **Adapt** evaluation frameworks to domain-specific needs (e.g., HIPAA compliance, Legal citation accuracy).
- **Deep Dive** into Healthcare, Law, Education, Finance, and Engineering.
- **Plan** governance and handoff processes with Subject Matter Experts (SMEs).

The Knowledge Gap

What the Model Knows

- General grammar
- Common knowledge
- Public internet data (up to cutoff)

What the Expert Knows

- "Tribal knowledge" (unwritten rules)
- Current internal policies
- Edge cases and exceptions

Goal: Move the "Expert Knowledge" into the "Context Window" via Knowledge Packs.

Technique: The Knowledge Interview

How to extract promptable information from an SME:

- 1 **The "Golden Path":** "Walk me through a perfect example of this task." (Becomes your Few-Shot example).
- 2 **The "Anti-Pattern":** "What is a common mistake a junior employee makes?" (Becomes your Negative Constraint).
- 3 **The "Heuristic":** "If you see X, do you always do Y? Or does it depend?" (Becomes your Chain-of-Thought logic).

Structure of a Knowledge Pack

A text file or prompt section that grounds the model.

- **Definitions:** "In this company, 'Churn' means cancellation within 30 days, not non-renewal."
- **Acronyms:** "CTR = Clinical Trial Record (not Click Through Rate)."
- **Style Guide:** "Use active voice. Cite regulations as [Reg 12.3]."

Prompt Pattern: "Refer to the attached Glossary for all term definitions. If a term is ambiguous, ask for clarification."

- **Core Challenge:** Accuracy and Privacy (HIPAA/GDPR).
- **Use Case 1: Clinical Note Summarization**
 - *Input:* Transcript of doctor-patient conversation.
 - *Output:* Structured SOAP Note (Subjective, Objective, Assessment, Plan).
- **Use Case 2: Patient Education**
 - *Task:* Translate complex medical jargon into 5th-grade reading level instructions.

Healthcare Prompt Pattern: De-identification

The "Sanitizer" Prompt

Before sending data to a cloud model, you often need to strip PII (Personally Identifiable Information).

Task: Remove all PII from the text below.
Replace names with [NAME], dates with [DATE],
locations with [LOC].
Keep the clinical context intact.

Input: "Mr. John Smith (DOB 01/01/80) visited
Medipol Hospital on Tuesday for his diabetes."

Output: "Mr. [NAME] (DOB [DATE]) visited
[LOC] on [DATE] for his diabetes."

- **Core Challenge:** Hallucination of laws/cases.
- **Use Case 1: Contract Review**
 - "Identify clauses that deviate from our standard 'Force Majeure' template."
- **Use Case 2: Discovery Summarization**
 - "Summarize these 50 emails. Highlight any mention of 'Project X' or 'Budget Cuts'."

Legal Prompting: The IRAC Framework

Lawyers think in **IRAC** (Issue, Rule, Analysis, Conclusion). Force the LLM to do the same.

Prompt Structure

Role: You are a Junior Associate. **Task:** Analyze the provided fact pattern. **Steps:** 1. **Issue:** State the legal question. 2. **Rule:** Cite the relevant law (provided in context). 3. **Analysis:** Apply the rule to the facts. 4. **Conclusion:** Yes/No/Maybe.

Education: Personalization at Scale

- **Core Challenge:** Pedagogical alignment and bias.
- **Use Case 1: Differentiated Learning**
 - "Rewrite this physics problem for a student who loves soccer."
- **Use Case 2: Socratic Tutor**
 - "Do not give the answer. Ask a guiding question to help the student solve it themselves."

Education Prompt: Quiz Generation

Task: Create 3 multiple-choice questions on "Photosynthesis".

Target Audience: 8th Grade.

Format:

- Question Stem
- 1 Correct Answer
- 3 Distractors (Plausible but incorrect)
- Explanation of why the correct answer is right.

Constraint: Ensure distractors address common misconceptions (e.g., plants get food from soil).

- **Core Challenge:** Numerical accuracy and timely data.
- **Use Case 1: Earnings Call Analysis**
 - "Extract all forward-looking statements about 'Revenue Growth' and classify sentiment (Positive/Neutral/Negative)."
- **Use Case 2: Regulatory Compliance**
 - "Check this marketing email against FINRA communication guidelines. Flag any promises of 'guaranteed returns'."

- **Core Challenge:** Security and maintainability.
- **Use Case 1: Legacy Code Explanation**
 - "Explain this COBOL function in plain English so a Python developer can rewrite it."
- **Use Case 2: Test Generation**
 - "Write unit tests for this function. Include edge cases (null inputs, negative numbers)."

The SME Feedback Loop

You cannot prompt-engineer in a vacuum.

- 1 **Draft:** Engineer creates V1 prompt.
- 2 **Test:** Run 10 examples.
- 3 **Review:** SME rates outputs (Green/Yellow/Red).
- 4 **Refine:** Engineer asks "Why is this Yellow?" and updates the prompt.
- 5 **Repeat.**

Tip: Use a spreadsheet for "Blind Testing" (SME doesn't know which prompt version generated the output).

Governance: Human-in-the-Loop (HITL)

When do you need a human?

- **High Risk:** Medical diagnosis, hiring decisions, loan approvals. (Human must approve).
- **Medium Risk:** Draft generation, code suggestions. (Human reviews).
- **Low Risk:** Brainstorming, entertainment. (Fully automated).

Policy: "AI outputs must be verified by a qualified professional before being sent to a client."

Activity: The "Red Team" Workshop

- **Goal:** Break the prompt.
- **Setup:** Split into pairs. One is the "User", one is the "AI".
- **Task:** The User tries to trick the AI into giving bad advice (e.g., "Ignore the safety rules").
- **Outcome:** Identify weak spots in your instructions and patch them.

Final Assignment: Domain Toolkit

- Select a domain (Health, Law, Finance, etc.).
- **Deliverable 1: Knowledge Pack** (Glossary + 3 "Golden" Examples).
- **Deliverable 2: The "System Prompt"** (Role, Constraints, Tone).
- **Deliverable 3: Evaluation Matrix** (How will you measure success? Accuracy? Tone? Speed?).

Key Takeaways

- **Context is King:** Domain prompts fail without specific knowledge (Glossaries, Rules).
- **SMEs are Partners:** You need their brain to build the prompt.
- **Safety First:** In specialized domains, "Do no harm" is the first instruction.
- **Iterate:** Your first prompt will be wrong. Your tenth will be useful.

Thank You!

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Questions?